CSCI 1470 Spring26 Final Project: EchoVision
Semantic-Driven Cross-Modal Image Synthesis from Music

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***Abstract*—We present EchoVision, a semantic-driven cross-modal generation system that translates music into images and text descriptions. Our approach introduces a shared label space as an intermediate representation, enabling interpretable and controllable mapping between audio and visual modalities. We train a convolutional neural network (CNN) to predict semantic tags from Mel spectrograms derived from audio clips, and use these labels to condition a Stable Diffusion model for image synthesis as well as a large language model (LLM) for explanation generation. Due to limited data availability, our model achieves a validation micro-F1 score of 0.075, with clear signs of overfitting. Despite limited quantitative performance, qualitative results demonstrate coherent cross-modal behavior. We further analyze challenges such as label sparsity, data scarcity, and error propagation, and discuss future directions for improving semantic alignment and model robustness.**

GitHub Repository:
https://github.com/Jing-Xu1223/EchoVision

I. Introduction

Music evokes vivid emotions, memories, and even imagined scenes, yet translating sound into visual representations remains a challenging computational problem. Humans naturally associate music with imagery—for example, melancholic jazz may evoke a rainy urban night, while upbeat electronic music may suggest vibrant, fast-paced environments. Capturing this intuitive cross-modal mapping in a computational framework, however, is non-trivial.

Recent advances in deep learning have enabled significant progress in unimodal tasks such as audio classification and image generation. However, bridging these modalities introduces several challenges. First, audio signals are inherently temporal and high-dimensional, making semantic interpretation difficult. Second, the relationship between sound and visual imagery is highly subjective, with no single ground-truth mapping. Finally, existing approaches often rely on end-to-end models, which lack interpretability and offer limited control over generated outputs.

In this project, we propose **EchoVision**, a semantic-driven cross-modal generation framework that transforms music into semantically aligned images and textual descriptions. Instead of directly mapping audio to images, we introduce a shared semantic label space (e.g., *melancholic*, *orchestral*, *energetic*) as an intermediate representation. This design improves interpretability, enables modular integration of different models, and allows more controllable generation.

Our approach combines three key components: (1) a convolutional neural network (CNN) for predicting semantic labels from audio spectrograms, (2) a diffusion-based generative model for synthesizing images conditioned on these labels, and (3) a large language model (LLM) for producing human-readable explanations. This pipeline allows us to explicitly model the relationship between sound, semantics, and visual output.

Potential applications of this work extend beyond academic interest. For example, semantic-driven music visualization can enhance user experience in music streaming platforms, enable adaptive visual interfaces, and support creative tools for artists and designers. More broadly, this framework contributes to the development of interpretable multimodal systems.

Our main contributions include:

- A semantic-driven pipeline for music-to-image generation
- Integration of CNN-based audio understanding with diffusion and language models
- Empirical analysis of key challenges, including label sparsity, data limitations, and error propagation

A. Motivation

We are inspired by the human ability to associate music with visual imagery and atmosphere. Rather than treating cross-modal generation as a purely black-box problem, we aim to design a system that reflects this reasoning process through an interpretable intermediate representation.

B. Goals

- **Base Goal:** Build a functional end-to-end pipeline that maps music to both images and textual descriptions through a shared semantic representation.
- **Target Goal:** Learn a meaningful semantic label space that captures key musical attributes (e.g., mood, genre, texture) and enables consistent cross-modal alignment between audio, image, and language.
- **Extended Goal:** Improve system robustness and usability through larger-scale data training and potential deployment as an interactive web-based application.

II. Literature Review

Our work builds on three related areas: audio representation learning, multimodal semantic alignment, and text-conditioned image generation.

A. Audio Representation Learning

Music and environmental audio are commonly represented as time-frequency features such as Mel spectrograms or log-Mel spectrograms. These representations convert an audio waveform into a two-dimensional structure that can be processed similarly to an image, making convolutional neural networks (CNNs) a natural choice for audio tagging and classification. Prior work such as PANNs demonstrates the effectiveness of large-scale CNN-based audio models for audio pattern recognition. Kong et al. propose pretrained audio neural networks trained on AudioSet and show that these models can transfer to other audio-related tasks [1]. This supports our decision to treat Mel spectrograms as visual-like inputs and train a CNN for multi-label music tagging.

In our project, we do not attempt to train a large pretrained audio model from scratch. Instead, we build a smaller custom CNN tailored to the MusicCaps label space. This design choice keeps the project computationally feasible while still allowing us to study how well spectrogram-based learning can recover semantic music descriptors.

B. Music Captioning and Semantic Supervision

Music-to-language datasets provide an important bridge between raw audio and human-interpretable meaning. MusicCaps is particularly relevant because each music clip is paired with an aspect list and a free-text caption written by musicians. The dataset description states that MusicCaps contains "5,521 music examples" with aspect lists and captions written by musicians [2]. This makes it well-suited for our semantic-label approach, since the aspect tags can be treated as weak supervision for concepts such as mood, genre, instrumentation, and texture.

Recent work on music captioning also highlights the challenge of limited music-language data. LP-MusicCaps, for example, explicitly addresses the scarcity of music-caption pairs by using large language models to generate pseudo captions from large-scale tag datasets [3]. This is directly relevant to our project because our results show that limited data and sparse labels are major bottlenecks for robust music-to-label prediction.

C. Multimodal Semantic Alignment

A central motivation for EchoVision is the idea that different modalities can be connected through a shared semantic representation. CLIP demonstrated that image and text representations can be aligned by training on large-scale image-caption pairs. Radford et al. argue that learning from natural language supervision provides a more flexible alternative to fixed category labels and allows models to reference visual concepts through text [4]. This idea informs our use of human-readable semantic labels as an intermediate layer between music and vision.

AudioCLIP extends this line of work by incorporating audio into the CLIP framework, creating a tri-modal system involving audio, image, and text [5]. While our project does not train a contrastive tri-modal embedding model, it shares a similar goal: connecting audio, visual, and language modalities through a common semantic interface. Our approach differs by making the intermediate representation explicit. Instead of learning a fully latent shared embedding, we use predicted labels such as *melancholic*, *orchestral*, or *energetic*, which makes the pipeline more interpretable but also more vulnerable to upstream classification errors.

D. Text-Conditioned Image Generation

For the image generation component, our work builds on diffusion-based text-to-image synthesis. Latent Diffusion Models reduce the computational cost of diffusion by operating in the latent space of pretrained autoencoders while retaining high visual quality. Rombach et al. describe latent diffusion as enabling high-quality and flexible image synthesis with conditioning inputs such as text [6]. This motivates our use of Stable Diffusion as the image generation backbone.

In EchoVision, the diffusion model is not directly conditioned on raw audio. Instead, predicted music labels are first converted into a natural language prompt. This modular design allows us to use the strengths of text-to-image diffusion while keeping the audio understanding component interpretable. However, it also introduces prompt sensitivity: small changes in predicted labels can lead to substantial changes in the generated image.

E. Position of Our Work

Compared with fully end-to-end audio-to-image systems, EchoVision emphasizes interpretability and modularity. The system first predicts semantic labels from music, then uses these labels to generate images and text explanations. This makes the model behavior easier to inspect and explain. At the same time, the semantic layer introduces a compounding error problem: if the audio classifier misses important labels, both the generated prompt and final image may drift away from the original music. Our project therefore studies not only whether music can be translated into visual content, but also how an explicit semantic bridge shapes the strengths and weaknesses of cross-modal generation.

III. Methodology

EchoVision is designed as a modular cross-modal generation pipeline. Rather than directly mapping raw audio to images, we first convert music into an interpretable semantic representation, then use that representation to drive both image generation and text explanation. This section describes the dataset, preprocessing pipeline, model architecture, training objective, and evaluation strategy.

A. Dataset

We use the MusicCaps dataset, which contains 5,521 music examples. Each example is associated with an English aspect list and a free-text caption written by musicians. The aspect lists describe musical attributes such as mood, genre, instrumentation, rhythm, vocal quality, and texture, making MusicCaps well-suited for semantic music understanding. In

our project, these aspect lists serve as weak supervision for the multi-label prediction task.

Although the full MusicCaps dataset contains 5,521 annotated clips, practical access limitations reduced the usable subset. Some YouTube-linked clips were unavailable, region-restricted, or failed during downloading. As a result, our final training subset contained 623 clips, split into 499 training clips and 124 validation clips. This restricted data scale became one of the major constraints of our project and directly influenced both model performance and generalization.

Each clip in MusicCaps is approximately ten seconds long. For consistency, all audio files were converted to a common format before being passed into the model. We kept the problem formulation close to the original dataset structure by treating each clip as one training example with a set of associated semantic labels.

B. Audio Preprocessing

Raw audio waveforms are difficult to use directly in a small custom model because they are high-dimensional and highly temporal. We therefore transform each audio clip into a Mel spectrogram, a two-dimensional time-frequency representation of sound. This representation maps acoustic energy across time and frequency bands, making it visually similar to an image and compatible with convolutional neural networks.

Given an audio waveform $x(t)$, the preprocessing pipeline first computes a short-time Fourier transform (STFT), then maps the frequency axis onto the Mel scale. The resulting Mel spectrogram can be represented as:

$$\mathbf{S}_{mel} = \text{MelFilterBank}(|\text{STFT}(x)|^2) \tag{1}$$

where $\mathbf{S}_{mel}$ is the Mel spectrogram used as the input to the CNN. This design is consistent with prior work in audio tagging and pattern recognition, where log-Mel spectrograms and convolutional models have been widely used for large-scale audio classification tasks.

Figure 1 illustrates the transformation from waveform to Mel spectrogram and then to predicted semantic labels.

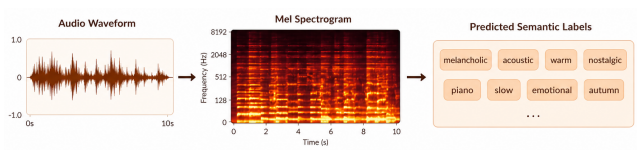

Fig. 1: Audio preprocessing and semantic label prediction. Raw audio is converted into a Mel spectrogram, which is then processed by a CNN classifier to predict semantic music labels.

C. Semantic Label Construction

The output of the CNN is a multi-label semantic vector rather than a single class. This choice reflects the nature of music: a clip can be simultaneously *melancholic*, *acoustic*, *slow*, and *emotional*. MusicCaps annotations often contain multiple descriptive tags per clip, so treating the task as multi-label classification is more appropriate than forcing each clip into one category.

Let K denote the number of unique labels in our vocabulary. Each training example is represented by a binary label vector:

$$\mathbf{y} \in \{0,1\}^K \tag{2}$$

where $y_i = 1$ indicates that the i-th semantic tag is present for the audio clip. In our final experiment, the label space contained 1,462 unique tags. While this large vocabulary preserves the richness of MusicCaps annotations, it also creates severe label sparsity. Many tags appear only a small number of times, making robust prediction difficult with only 623 usable training examples.

D. Music-to-Label CNN Classifier

The first learned component of EchoVision is a custom CNN classifier that maps Mel spectrograms to semantic labels. The model receives $\mathbf{S}_{mel}$ as input and produces a probability vector:

$$\hat{\mathbf{y}} = f_\theta(\mathbf{S}_{mel}) \tag{3}$$

where f_θ denotes the CNN with learnable parameters θ, and $\hat{\mathbf{y}} \in [0,1]^K$ contains predicted probabilities for all semantic labels.

The CNN consists of convolutional layers for local time-frequency feature extraction, nonlinear activations, pooling layers for dimensionality reduction, and fully connected layers for multi-label prediction. Conceptually, the convolutional filters learn patterns such as rhythmic density, harmonic texture, and spectral energy distribution. These features are then used to infer higher-level semantic descriptors such as mood, genre, and instrumentation.

Because this is a multi-label classification problem, we use a sigmoid activation for each label independently rather than a softmax over mutually exclusive classes:

$$\hat{y}_i = \sigma(z_i) = \frac{1}{1+e^{-z_i}} \tag{4}$$

where z_i is the raw output logit for label i. A threshold of 0.5 is applied during evaluation to convert predicted probabilities into binary label predictions.

E. Training Objective

We train the CNN using binary cross-entropy loss over all labels. For a single example, the loss is:

$$(\hat{\mathbf{y}}, \mathbf{y}) = -\sum_{i=1}^{K} \left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i) \right]. \tag{5}$$

This objective penalizes incorrect predictions independently for each semantic tag. We trained the model for 30 epochs using the Adam optimizer. The primary evaluation metric was micro-F1, which aggregates true positives, false positives, and false negatives across all labels before computing the F1

score. Micro-F1 is appropriate here because the task is highly imbalanced and many labels are rare.

F. Labels-to-Image Generation

The second stage of the pipeline converts predicted semantic labels into a visual output. Instead of training an image generator from scratch, we use a Stable Diffusion pipeline as the image synthesis backbone. Latent Diffusion Models such as Stable Diffusion have shown strong performance for text-conditioned image generation while reducing computational cost by performing diffusion in latent space rather than directly in pixel space.

To connect the CNN output to Stable Diffusion, we introduce a prompt engineering layer. This layer converts the predicted labels into a natural language prompt. For example, a predicted label set such as:

$$\{melancholic, jazz, night, urban\} \tag{6}$$

may be converted into a prompt such as:

> "A quiet rain-soaked city at night, with soft jazz atmosphere, neon reflections, and a melancholic mood."

This prompt is then passed to Stable Diffusion to synthesize an image. In this design, the semantic labels act as a controllable interface between the audio classifier and the generative image model. The benefit is interpretability: users can inspect the predicted labels and understand why a particular image was generated. The drawback is error propagation: if the CNN misses important tags, the prompt and generated image may drift away from the original music.

G. Labels-to-Text Explanation

In addition to image generation, EchoVision generates a short text explanation describing the relationship between the music and the visual output. The same predicted labels are passed into a Large Language Model (LLM), which produces a human-readable explanation of the inferred mood and scene. This explanation is not intended to be a separate ground-truth caption; instead, it serves as an interpretability layer for the user.

For example, given labels such as *melancholic*, *jazz*, *night*, and *urban*, the explanation might describe a quiet rainy city scene with soft jazz-like atmosphere. This reinforces the central idea of the project: the label space is not only a technical bridge between models, but also a human-readable bridge between music and visual interpretation.

H. Full Pipeline

The full EchoVision pipeline is shown in Figure 2. Audio clips are first transformed into Mel spectrograms and passed through the CNN classifier. The resulting semantic labels then branch into two downstream modules: prompt engineering for Stable Diffusion image generation and LLM-based text explanation.

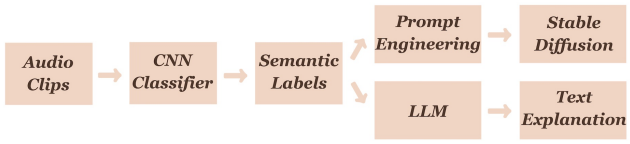

Fig. 2: Overview of the EchoVision pipeline. Audio clips are converted into Mel spectrograms, classified into semantic labels, and then used to generate both images and text explanations.

I. Evaluation Metrics

We evaluate the system at two levels. First, we evaluate the CNN classifier quantitatively using micro-F1 on the validation set. This measures how well the model predicts MusicCaps semantic labels. Second, we evaluate the end-to-end system qualitatively through generated image and text examples, as there is no single ground-truth image for a music clip.

To supplement qualitative inspection, we also conducted a small human evaluation in which participants rated how well generated images matched the corresponding music clips. Because music-to-image translation is inherently subjective, this evaluation is intended to measure perceived alignment rather than objective correctness.

IV. Results

We evaluate EchoVision at both the component level (music-to-label classification) and the full pipeline level (cross-modal generation quality). Due to the subjective nature of music-to-image translation, we combine quantitative metrics with qualitative and human evaluation.

A. Quantitative Results

The multi-label CNN was trained on 623 audio clips (499 training / 124 validation) over 30 epochs across a label space of 1,462 unique tags.

The best validation performance was achieved at epoch 3, with a micro-F1 score of 0.075 (threshold = 0.5). After this point, validation performance steadily declined while training loss continued to decrease, indicating clear overfitting.

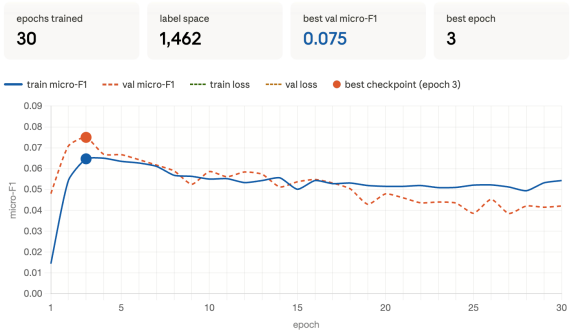

Fig. 3: Micro-F1 score over training epochs. Validation performance peaks early and declines, indicating severe overfitting due to limited data scale and high label sparsity.

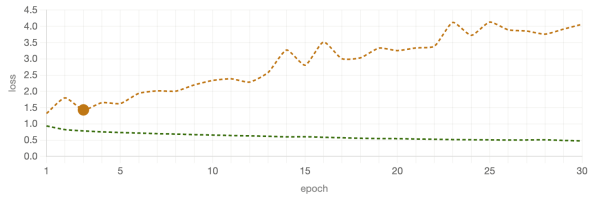

Fig. 4: Training and validation loss curves. Training loss continues to decrease while validation loss increases, further confirming overfitting.

The gap between training and validation curves highlights the difficulty of learning a high-dimensional multi-label space with limited data. The large label vocabulary (1,462 tags) combined with sparse annotations leads to poor generalization. This confirms that label sparsity is a primary bottleneck in music-to-label learning.

B. Qualitative Results

Despite limited quantitative performance, the full pipeline demonstrates coherent cross-modal behavior. Predicted labels are successfully transformed into prompts that generate visually and semantically consistent outputs.

Figure 5 shows representative end-to-end results. Each example includes the generated image, predicted labels, and LLM-generated description.

Fig. 5: End-to-end generation examples. The system produces visually distinct scenes aligned with predicted semantic labels.

We observe that broad emotional attributes are consistently preserved across modalities. For example:

- **Energetic / upbeat** labels produce bright, colorful urban scenes with strong lighting contrast.
- **Melancholic / jazz** labels generate darker, rain-soaked environments with soft lighting and atmospheric depth.
- **Classical / calm** labels result in structured, symmetrical indoor scenes with warm tones and minimal motion cues.

These patterns suggest that the semantic label space captures meaningful high-level attributes that transfer well into visual generation.

However, failure cases occur when key labels are missing or incorrectly predicted. In such cases, the generated image and text explanation drift away from the original musical content. This confirms that the quality of the semantic representation is the most critical factor for end-to-end performance.

C. Human Evaluation

To further assess perceptual alignment, we conducted a human evaluation study.

We distributed a questionnaire via the WeChat platform (Wenjuanxing) to approximately 50 participants. After data cleaning and removing incomplete responses, we obtained 47 valid participants. Each participant rated 5 audio-image pairs, resulting in a total of 235 ratings. Participants evaluated how well each generated image matched the corresponding music on a 5-point Likert scale (from strongly mismatched to strongly matched).

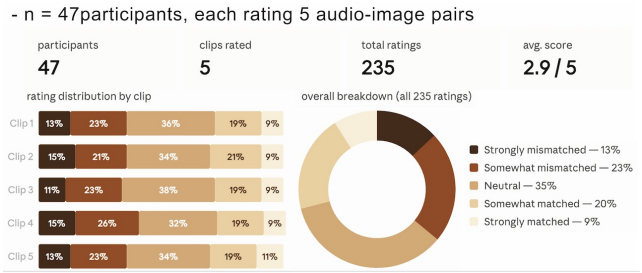

Fig. 6: Human evaluation results. Distribution of ratings across all audio-image pairs.

The average rating was 2.9 / 5. The distribution shows:

- 29% of ratings were *matched* or *strongly matched*
- 35% were *neutral*
- 36% were *mismatched* to varying degrees

These results indicate that while the system can produce convincing matches in some cases, performance is inconsistent across inputs. The relatively high proportion of neutral responses suggests that the model often captures general mood but lacks fine-grained alignment.

D. Goal Achievement

We successfully achieved our base and target goals:

- Built a complete end-to-end pipeline from music to image and text
- Demonstrated interpretable cross-modal generation via semantic labels

However, stretch goals were only partially achieved. In particular, bidirectional modeling (image $\rightarrow$ music) was not implemented due to time and data constraints. Additionally, limited dataset size prevented robust training of the multi-label classifier, which in turn constrained overall system performance.

Overall, the results highlight both the promise and limitations of semantic-driven cross-modal generation: while the framework enables interpretable and modular design, its effectiveness is highly dependent on the quality of the intermediate label representation.

V. Challenges

Our experimental results highlight several fundamental challenges in semantic-driven cross-modal generation.

A. Data Scarcity

Although MusicCaps provides 5,521 annotated examples, only 623 clips were usable in our setting due to access constraints. This limited scale significantly restricts the ability of the model to learn robust mappings from audio to high-dimensional semantic labels. Prior work has shown that large-scale pretraining is critical for stable audio representation learning, particularly in multi-label settings.

B. Label Sparsity

The label space in our experiments contains 1,462 unique tags, many of which appear only a few times. This creates a highly sparse supervision signal, making it difficult for the model to generalize. Similar challenges have been reported in music tagging literature, where long-tail label distributions degrade classification performance and require either label filtering or hierarchical modeling.

C. Overfitting Under Limited Data

As shown in our results, validation micro-F1 peaks early and then declines while training loss continues to decrease. This indicates that the model quickly memorizes training patterns but fails to generalize. This phenomenon is consistent with standard observations in deep learning when model capacity exceeds effective dataset size.

D. Error Propagation in Modular Pipelines

Our design explicitly introduces a semantic label layer to improve interpretability. However, this also creates a compounding error problem: inaccuracies in label prediction directly affect both downstream image generation and text explanation. This tradeoff between interpretability and robustness is a key limitation of modular architectures compared to fully end-to-end systems.

E. Prompt Sensitivity in Generative Models

Small variations in predicted labels can lead to large changes in generated images due to the sensitivity of text-to-image diffusion models. This instability has been widely observed in prompt-based generation systems and makes consistent output difficult to guarantee.

VI. Ethics

A. Subjectivity and Interpretation Bias

EchoVision operates in a domain where outputs are inherently subjective and open to interpretation. Unlike classification tasks with clear ground truth, there is no single "correct" image for a given piece of music. This introduces interpretation bias, where generated outputs reflect one possible interpretation among many.

B. Dataset Bias

The MusicCaps dataset is annotated by human experts, meaning that cultural, stylistic, and subjective biases are embedded in the semantic labels. The annotators are drawn from a limited cultural and musical background, which may influence how music is described (e.g., genre conventions, emotional interpretation). These biases propagate through the system and affect both generated images and textual descriptions.

C. Generative Model Risks

EchoVision relies on pretrained generative models such as Stable Diffusion, which have known limitations related to bias, training data provenance, and potential copyright concerns. Since these models are trained on large-scale web data, they may implicitly reproduce stylistic patterns or visual features associated with copyrighted material. In addition, they may reflect biases present in the underlying training distribution.

D. Misuse and Responsible Deployment

Generative systems capable of producing realistic content can be misused to create misleading or deceptive media. While our system is intended for creative and exploratory applications, similar pipelines could be adapted for unintended purposes.

We therefore emphasize that EchoVision should be used as a tool for creative visualization and research exploration, rather than as an authoritative system for interpreting music or generating factual representations.

VII. Reflection

A. Overall Assessment

We successfully implemented a complete end-to-end pipeline that transforms music into both images and textual descriptions. While quantitative performance of the music-to-label classifier was limited (micro-F1 = 0.075), the system demonstrates meaningful cross-modal behavior in qualitative outputs. This highlights a key insight: even imperfect semantic representations can support coherent downstream generation when they capture high-level attributes such as mood and texture. This reflects the exploratory nature of the project, where understanding the limitations of semantic alignment was as valuable as achieving strong performance.

B. Evolution and Pivots

Our project evolved significantly over time. Initially, we explored multiple directions, including direct multimodal generation and alternative input modalities. However, we ultimately converged on a semantic-driven approach, introducing an explicit label space as an intermediate representation.

This pivot was motivated by the need for interpretability and modularity. By separating the system into distinct components (audio understanding, image generation, and language explanation), we were able to analyze each stage independently. At the same time, this design introduced new challenges, particularly error propagation across modules.

C. Future Work

Several directions could substantially improve system performance.

First, scaling the dataset would likely provide the largest gain. Increasing the number of training samples and filtering low-frequency labels could reduce sparsity and improve generalization. Prior work suggests that large-scale pretraining is essential for robust audio tagging.

Second, improving the semantic representation is critical. Instead of treating labels as independent binary variables, future work could explore structured representations, such as hierarchical label spaces or learned embeddings, to better capture relationships between musical attributes.

Third, evaluation methods should be expanded. Since music-to-image generation lacks ground truth, incorporating larger-scale human evaluation or preference learning could provide more reliable assessment.

Finally, extending the system to bidirectional modeling (image $\rightarrow$ music) and deploying a real-time interactive interface would further validate the framework and broaden its applicability.

D. Key Takeaways

This project highlights several important lessons:

- Interpretability introduces tradeoffs: explicit semantic representations improve transparency but increase sensitivity to upstream errors.
- Multimodal learning is highly data-dependent: performance is constrained not only by model design but also by dataset scale and label quality.
- High-level semantics transfer well across modalities: even simple labels such as mood and genre can drive coherent cross-modal generation.

VIII. Conclusion

This project highlights the strengths and limitations of semantic-driven cross-modal generation.

Our results show that high-level semantic representations, such as mood and texture, can serve as an effective bridge between audio, vision, and language. Even with limited quantitative performance, the system is able to produce qualitatively coherent outputs, suggesting that coarse semantic alignment transfers well across modalities.

At the same time, our findings emphasize that the quality of the intermediate representation is critical. Label sparsity, limited data, and prediction errors significantly impact downstream generation, leading to instability and output drift. This reveals a key tradeoff: while explicit semantic layers improve interpretability, they also amplify error propagation in modular pipelines.

Overall, EchoVision demonstrates that interpretable multimodal systems are feasible, but highly dependent on data scale and representation quality. Future progress in this area will likely require better semantic representations, larger and more diverse datasets, and more robust evaluation frameworks.

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Finally, we would like to thank our collaborators and the people who accompanied us throughout this project. Even though the system is not perfect, this experience has been both meaningful and inspiring, and we learned a great deal from exploring this ambitious idea together.

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